

INDITEX UDC

Cátedra de IA en Algoritmos Verdes

MICROCREDENCIAL ALGORITMOS VERDES PARA LA IA: DISEÑO E IMPLEMENTACIÓN

Tema: AI, Graphics & Visual Computing: Usos de la IA para la Reconstrucción 3D. Generación de Datos Sintéticos.

Sesión: Lunes 6 de octubre de 2025

Ponente: José A. Iglesias Guitián



UNIVERSIDADE DA CORUÑA



Financiado por la
Unión Europea
NextGenerationEU



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DE ESPAÑA
MINISTERIO
PARA LA TRANSFORMACION DIGITAL
Y DE LA FUNCIÓN PÚBLICA



20
26

Session Outline

1. Photogrammetry and Computer Vision Fundamentals
2. A gentle introduction to Novel View Synthesis (NVS)
3. **Synthetic image data generation for AI consumption**

Synthetic Data in AI

Where does most of your training data come from today?

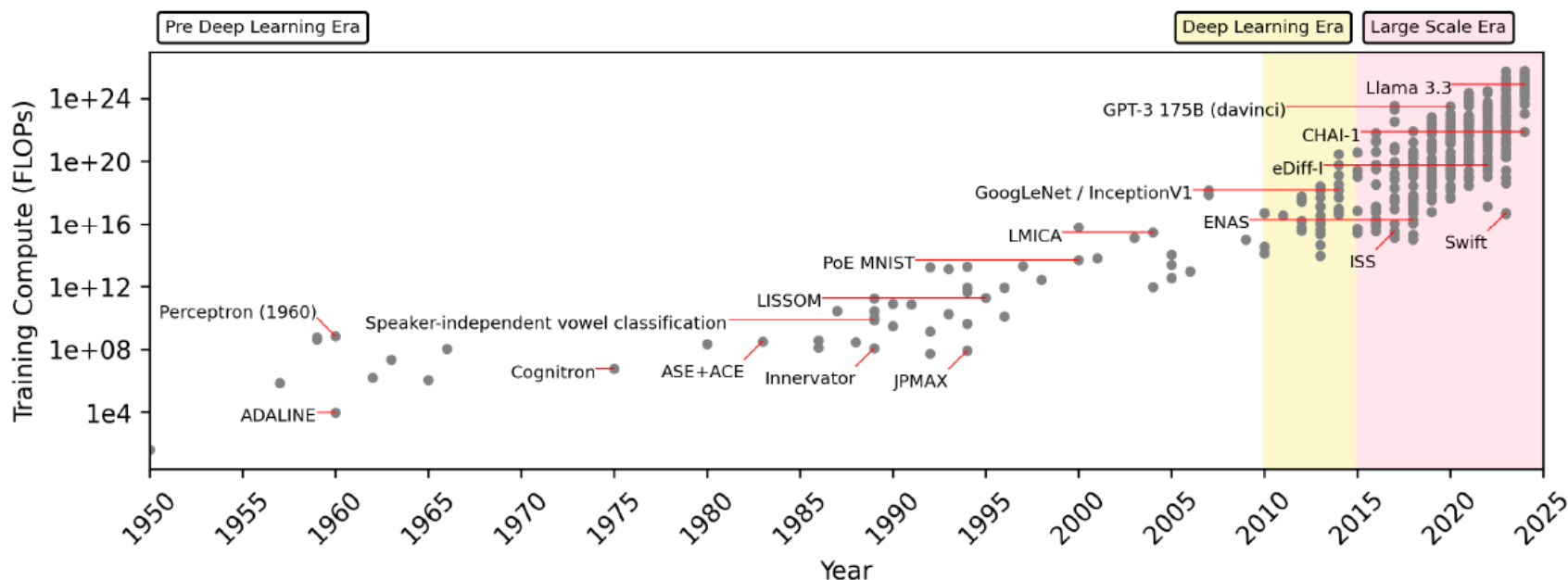
- Public benchmark datasets
 - e.g. ImageNet, COCO, Cityscapes, KITTI, MNIST, etc.
- Company-owned / in-house data collection
 - customer data, sensor recordings, logs
- Crowdsourced labeling services
 - Mechanical Turk, Scale AI, Appen
- Web scraping / weakly labeled data
 - images from the internet, videos, social media
- Purchased/licensed datasets
 - especially in healthcare, finance, autonomous driving

Challenges of Real-world Datasets

- **High labeling cost**
 - Manual annotation (bounding boxes, masks, segmentation, keypoints) is time-consuming and expensive.
 - Specialized domains (e.g. medical imaging, autonomous driving) require expert annotators.
- **Data scarcity for rare events**
 - Edge cases (e.g. car crashes, rare diseases) are crucial for robust AI, but too rare to collect at scale.
- **Privacy & legal concerns**
 - Facial recognition, health data, surveillance footage → subject to strict regulations (GDPR, HIPAA, etc.).
 - Often need anonymization or synthetic alternatives.
- **Bias and imbalance**
 - Datasets often overrepresent some demographics, geographies, or conditions.
 - Leads to biased AI models.
- **Safety & feasibility limits**
 - Some scenarios are unsafe to capture (e.g. car accidents, explosions, hazardous environments).
 - Others are logistically impossible to reproduce repeatedly.
- **Domain variability and lack of control**
 - Lighting, camera settings, weather, occlusions — hard to control in natural data collection.
 - Makes it tricky to systematically test robustness.
- **Scale & storage costs**
 - Collecting, storing, and managing millions of high-resolution images or sensor streams is resource-heavy.
- **Annotation errors & inconsistency**
 - Human labelers make mistakes or apply inconsistent criteria.
 - Quality control becomes another costly step.

Challenges of Real-world Datasets (Green AI perspective)

- The development and deployment of AI technologies depend on the ability to **scale infrastructure**.
 - Frontier AI applications resulted in rising **financial, environmental, and social costs**.
 - The **“data cliff”** is considered a soon-impending **barrier to further scaling of AI models**
 - Particularly for LLMs, as publicly available, human-generated data on the Internet is running out [Villalobos 2022].

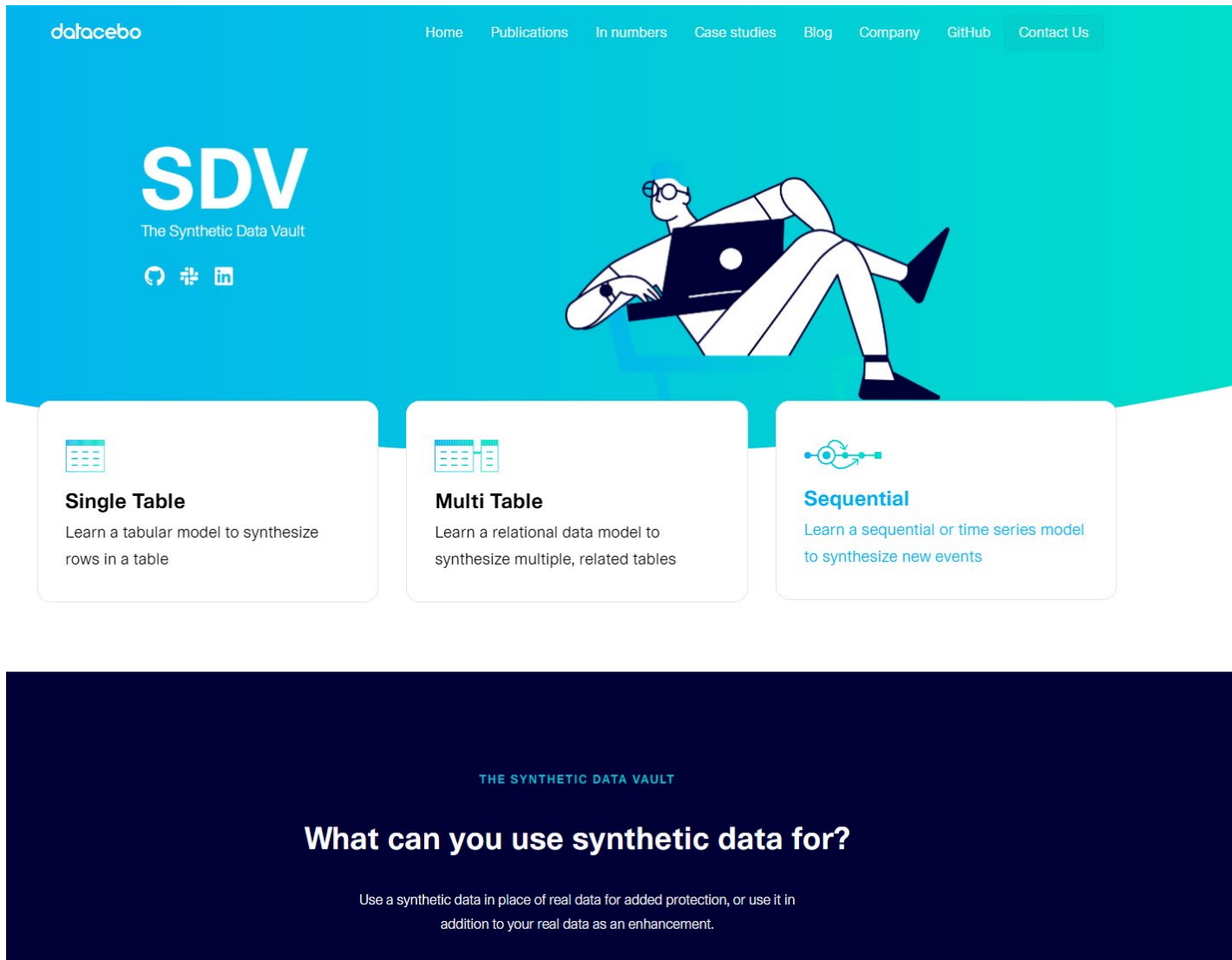


Source: Bhardwaj et al. Limits to AI Growth: The Ecological and Social Consequences of Scaling

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- Company-owned / in-house data collection
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 - images from the internet, videos, social media
- Purchased/licensed datasets
 - especially in healthcare, finance, autonomous driving
- (Rarely) **Synthetic or simulated data**
 - still a growing practice, not the norm yet

Example: Open-source data generation tools - Synthetic Data Vault



The screenshot shows the homepage of the Synthetic Data Vault (SDV) website. The header features the 'dfacebo' logo and navigation links for Home, Publications, In numbers, Case studies, Blog, Company, GitHub, and Contact Us. The main section has a large 'SDV' title with the subtitle 'The Synthetic Data Vault' and social media icons. An illustration of a person using a laptop is positioned to the right. Below this, three cards describe the tool's capabilities: 'Single Table' (learning a tabular model), 'Multi Table' (learning a relational data model), and 'Sequential' (learning a sequential or time series model). The footer section, titled 'THE SYNTHETIC DATA VAULT', asks 'What can you use synthetic data for?' and provides a brief explanation of its use.

dfacebo

Home Publications In numbers Case studies Blog Company GitHub Contact Us

SDV

The Synthetic Data Vault

Single Table

Learn a tabular model to synthesize rows in a table

Multi Table

Learn a relational data model to synthesize multiple, related tables

Sequential

Learn a sequential or time series model to synthesize new events

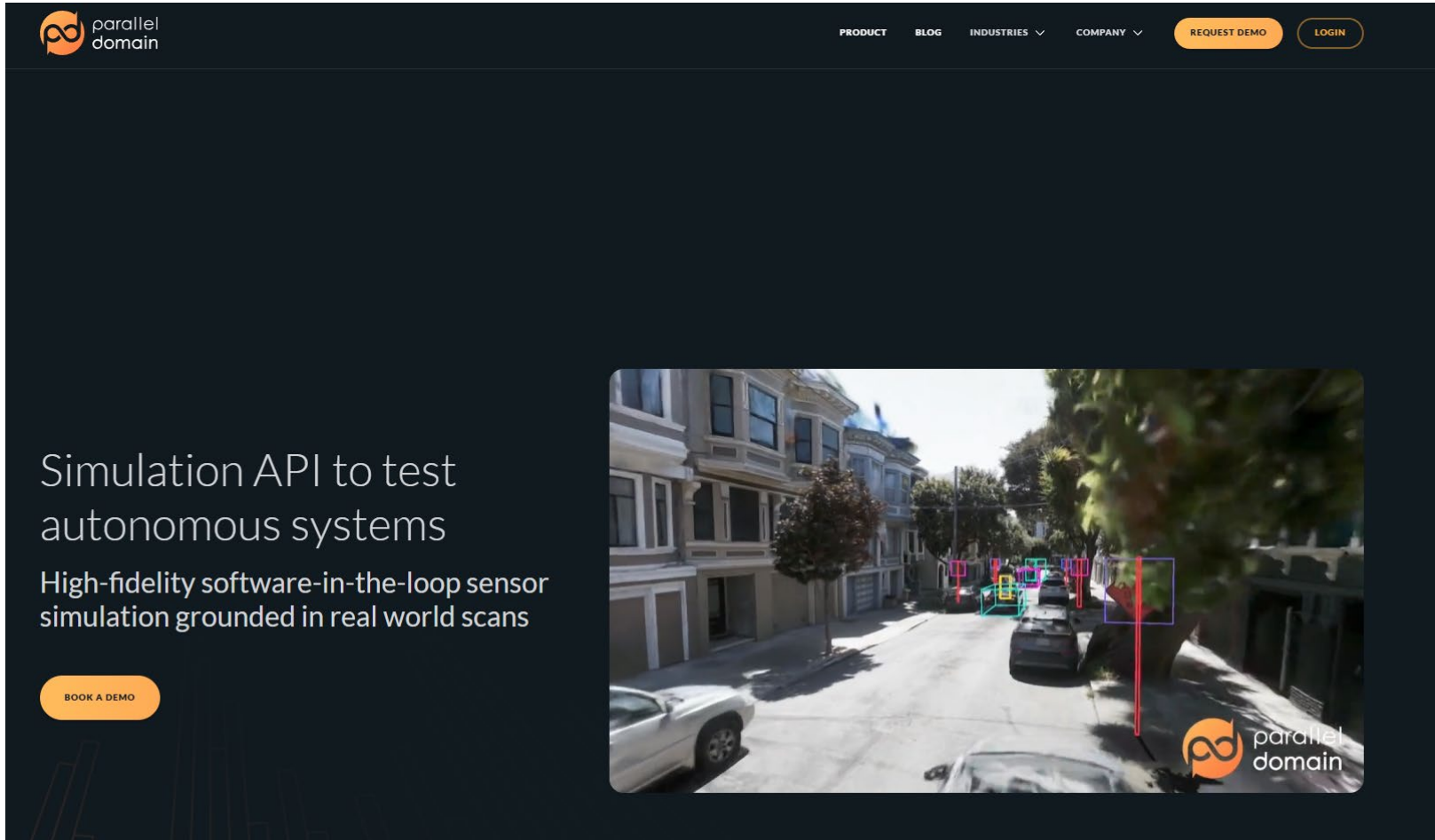
THE SYNTHETIC DATA VAULT

What can you use synthetic data for?

Use a synthetic data in place of real data for added protection, or use it in addition to your real data as an enhancement.

Source: <https://sdv.dev/>

Companies working as providers of Synthetic Data



The screenshot shows the Parallel Domain website. The top navigation bar includes the logo, links for PRODUCT, BLOG, INDUSTRIES, and COMPANY, and buttons for REQUEST DEMO and LOGIN. The main content area features the text "Simulation API to test autonomous systems" and "High-fidelity software-in-the-loop sensor simulation grounded in real world scans". A video player shows a street scene with a car and various colored bounding boxes (red, green, blue, yellow) indicating object detection. A "BOOK A DEMO" button is located below the text. The Parallel Domain logo is also visible in the bottom right corner of the video player.

parallel domain

PRODUCT BLOG INDUSTRIES COMPANY REQUEST DEMO LOGIN

Simulation API to test autonomous systems

High-fidelity software-in-the-loop sensor simulation grounded in real world scans

BOOK A DEMO

parallel domain

Source: <https://paralleldomain.com/>

How Synthetic Data helps address Real-world Data Challenges ...

- **High labeling cost**
 - Annotations (bounding boxes, masks, depth, keypoints) are generated automatically during rendering, eliminating manual labeling costs — even for complex domains like medical imaging or driving.
- **Data scarcity for rare events**
 - Rare or dangerous scenarios (accidents, rare diseases, system failures) can be generated on demand, ensuring robust coverage of edge cases.
- **Privacy & legal concerns**
 - Synthetic data contains no personally identifiable information (PII) and can mimic real distributions without exposing private records, making it GDPR/HIPAA compliant by design.
- **Bias and imbalance**
 - Data can be balanced programmatically (e.g. equal demographics, varied geographies) to mitigate bias and improve model fairness.
- **Safety & feasibility limits**
 - Dangerous or infeasible scenarios (crashes, explosions, extreme weather) can be safely simulated and repeated at scale.
- **Domain variability and lack of control**
 - Scene parameters (lighting, weather, camera angles, occlusion) can be controlled and systematically varied to test model robustness under different conditions.
- **Scale & storage costs**
 - Synthetic pipelines can generate virtually unlimited data at lower marginal cost, without the logistical overhead of large-scale real-world collection.
- **Annotation errors & inconsistency**
 - Labels generated programmatically are perfectly consistent and error-free, avoiding human annotation mistakes.

Source: Javier Taibo @ Course on Modeling at UDC (Chapter 10. Photogrammetry)

Synthetic datasets

- Types:
 - Synthetic tabular/text data. Not covered in this class, but [here](#):
 - “Mitigating Class Imbalance in Tabular Data through Neural Network-based Synthetic Data Generation: A Comprehensive Survey and Library”. TechRxiv 2025.
 - **Synthetic images**: generated through rules, simulations, or generative models.
 - e.g. GAN-generated faces, procedural driving scenes, synthetic X-ray images ...

Methods for Generating Synthetic Images

- Computer Graphics (CG) / Simulation engines:
 - Unity, Unreal Engine, Blender, etc. Use of 3D assets, rendering pipelines.
 - Annotation comes “for free” (bounding boxes, segmentation masks, depth maps, etc.)
- Programmatic / Procedural generation
 - domain-specific rules, often used in robotics.

Synthetic Data Generation through Computer Graphics

3D geometry reconstruction

Detail modeling and local shading

Texturing and appearance modeling

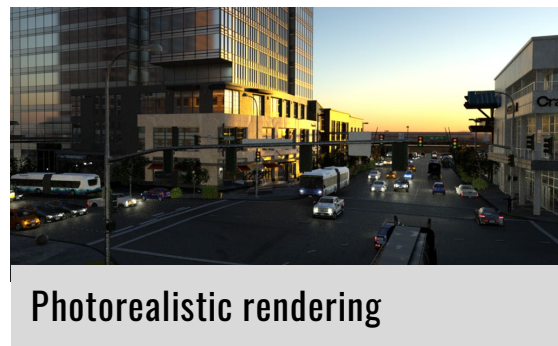
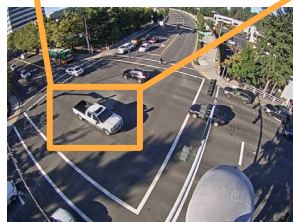
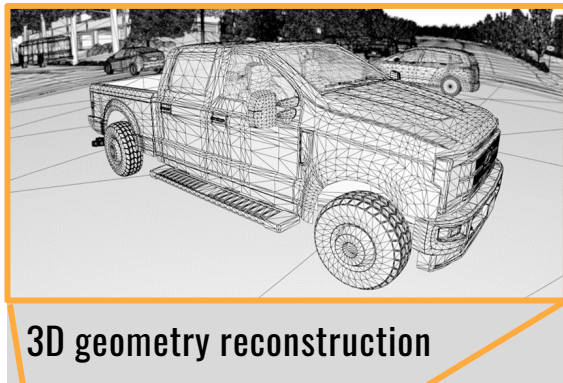


Virtual simulation

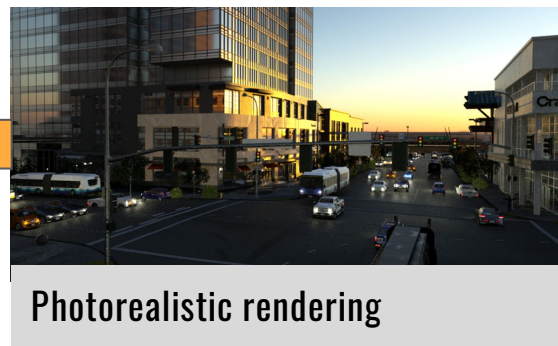
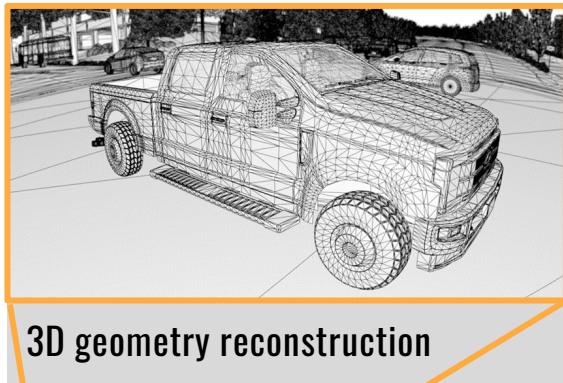
Photorealistic rendering

Appearance modeling and lighting

Synthetic Data Generation through Computer Graphics



Synthetic Data Generation through Computer Graphics



Synthetic Data Generation through Computer Graphics



Global illumination

Different lighting conditions



What is the Synth-to-Real Domain Gap Problem?

- **Definition:** Models trained purely on synthetic data often perform worse on real-world data.
- Synthetic images used to differ in texture, lighting, noise, and complexity compared to real data. However, this is changing with the latest generative AI and diffusion models.



Source: © NVIDIA Drive Sim DRIVE Sim creates a digital twin of the real world to provide a realistic driving environment.

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Source: © helm.ai Generative AI with Video Editing Capabilities for Autonomous Driving Development

Synth-to-Real Domain Gap Problem

- Causes of the domain gap:
 - Rendering realism:
 - technical limitations, artifacts, smooth surfaces, “too clean” scenes.
 - Physics mismatch:
 - synthetic environments may not fully simulate real-world dynamics.
 - Sensor differences:
 - synthetic images often miss real camera noise, lens distortion, motion blur.
 - Distribution mismatch:
 - differences in frequency of objects, textures, demographics.
- Consequences:
 - **Drop in model accuracy** when moving from training (synthetic) → testing (real).
 - Need for **real data fine-tuning**.
 - Limits the adoption of synthetic pipelines.

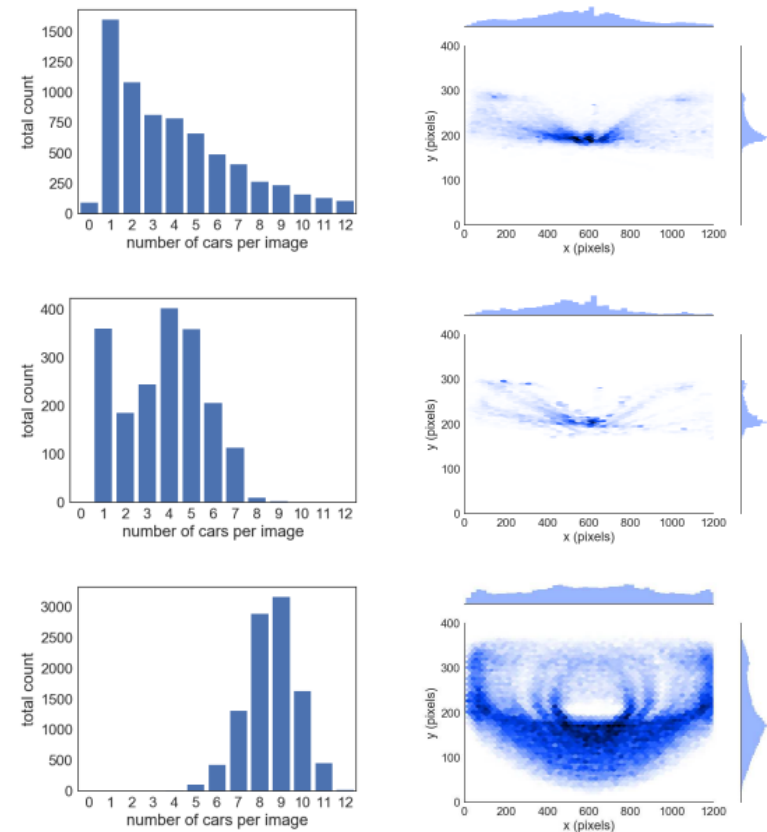


Figure 3. Statistics of the KITTI (top), virtual KITTI (middle), and our DR dataset (bottom). Shown are the distributions of the number of cars per image (left) and car centroid location (right). Note that with DR it is much easier to generate data with a wide distribution.

Synth-to-Real. Bridging the Gap

- Domain Randomization:
 - Over-diversify synthetic data (lighting, textures, noise).
- Domain Adaptation:
 - Use style transfer (e.g., CycleGAN) to make synthetic look more realistic.
- Hybrid Training:
 - Pre-train on synthetic, fine-tune with a smaller real dataset.
- Physics-based rendering:
 - Push synthetic realism closer to reality.

Synth-to-Real. Bridging the Gap

- **Domain Randomization:**

- Over-diversify synthetic data (lighting, textures, noise).



Figure 2. Sample images from Virtual KITTI (first row), and our DR approach (second row). Note that our images are easier to create (because of their lack of fidelity) and yet contain more variety to force the deep neural network to focus on the structure of the objects of interest.

Source: Tremblay et al. Training Deep Networks with Synthetic Data: Bridging the Reality Gap by Domain Randomization, CVPR 2018

Synth-to-Real. Bridging the Gap

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- Use style transfer (e.g., CycleGAN) to make synthetic look more realistic.

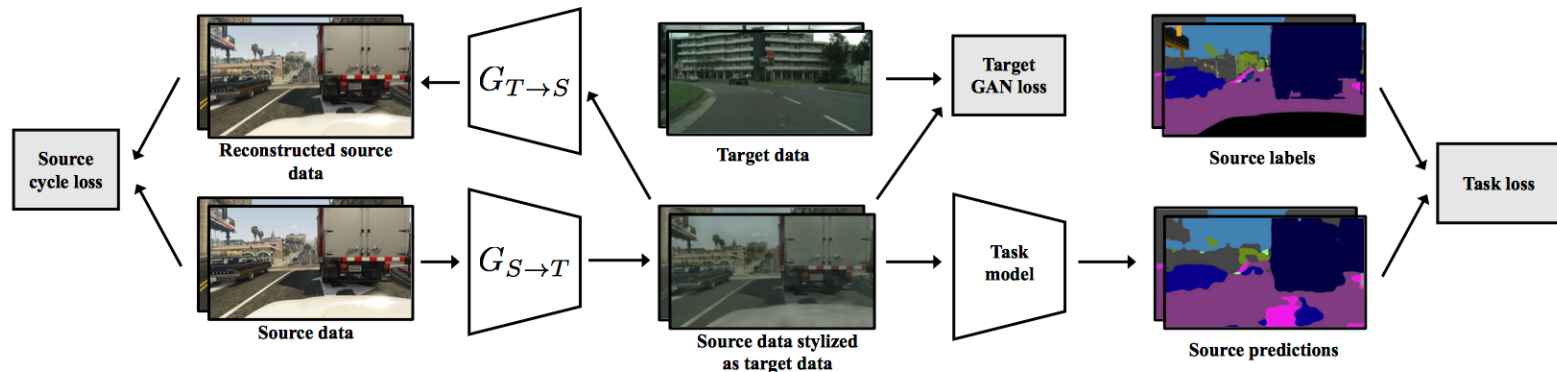


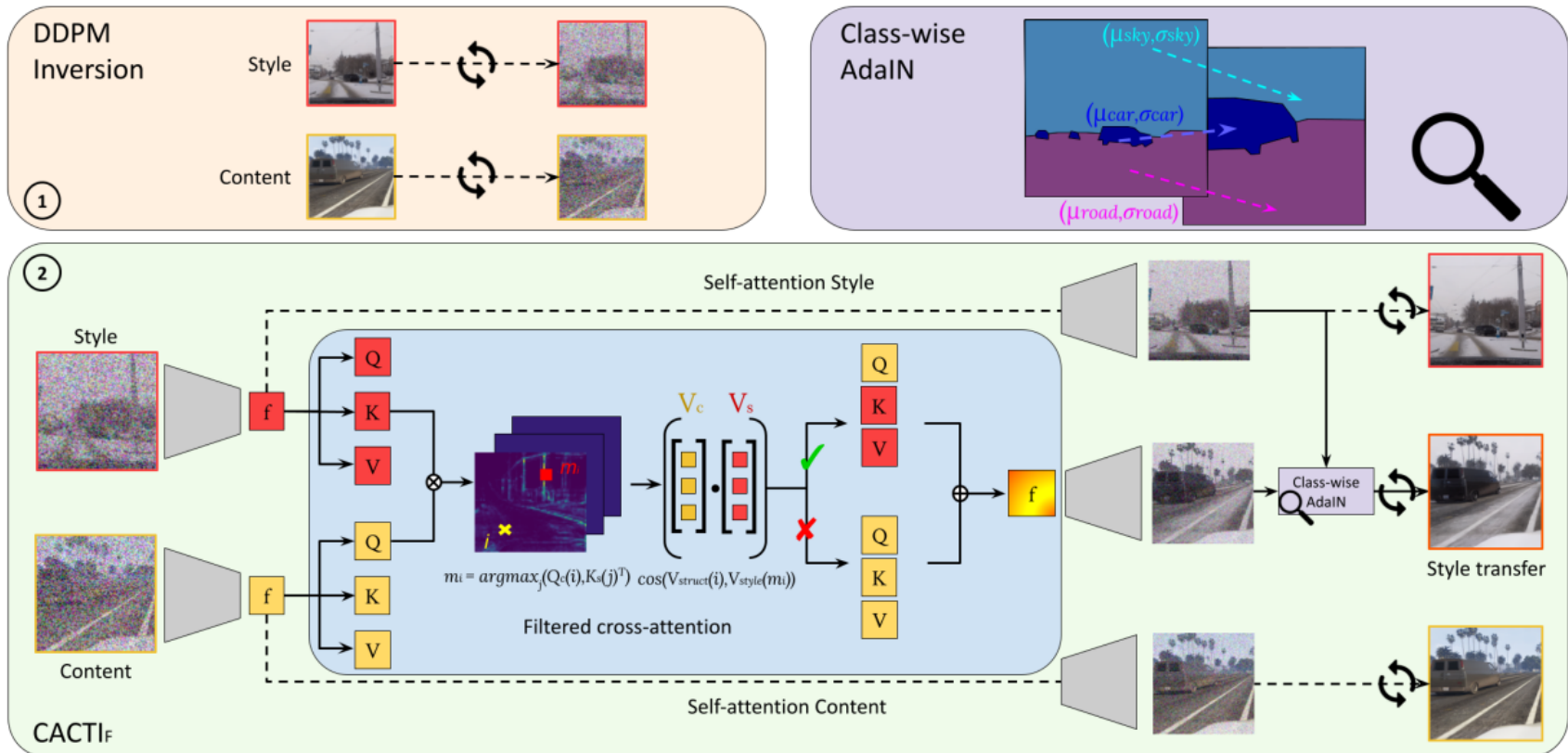
Figure 2: Cycle-consistent adversarial adaptation of pixel-space inputs. By directly remapping source training data into the target domain, we remove the low-level differences between the domains, ensuring that our task model is well-conditioned on target data. Our model also includes a target cycle loss and the source GAN loss which we omit here for clarity.

Source: Hoffman et al. CyCADA: Cycle-Consistent Adversarial Domain Adaptation, ICML 2018

Synth-to-Real. Bridging the Gap

- **Domain Adaptation:**

- Use style transfer (e.g., Diffusion Model) to make synthetic look more realistic.

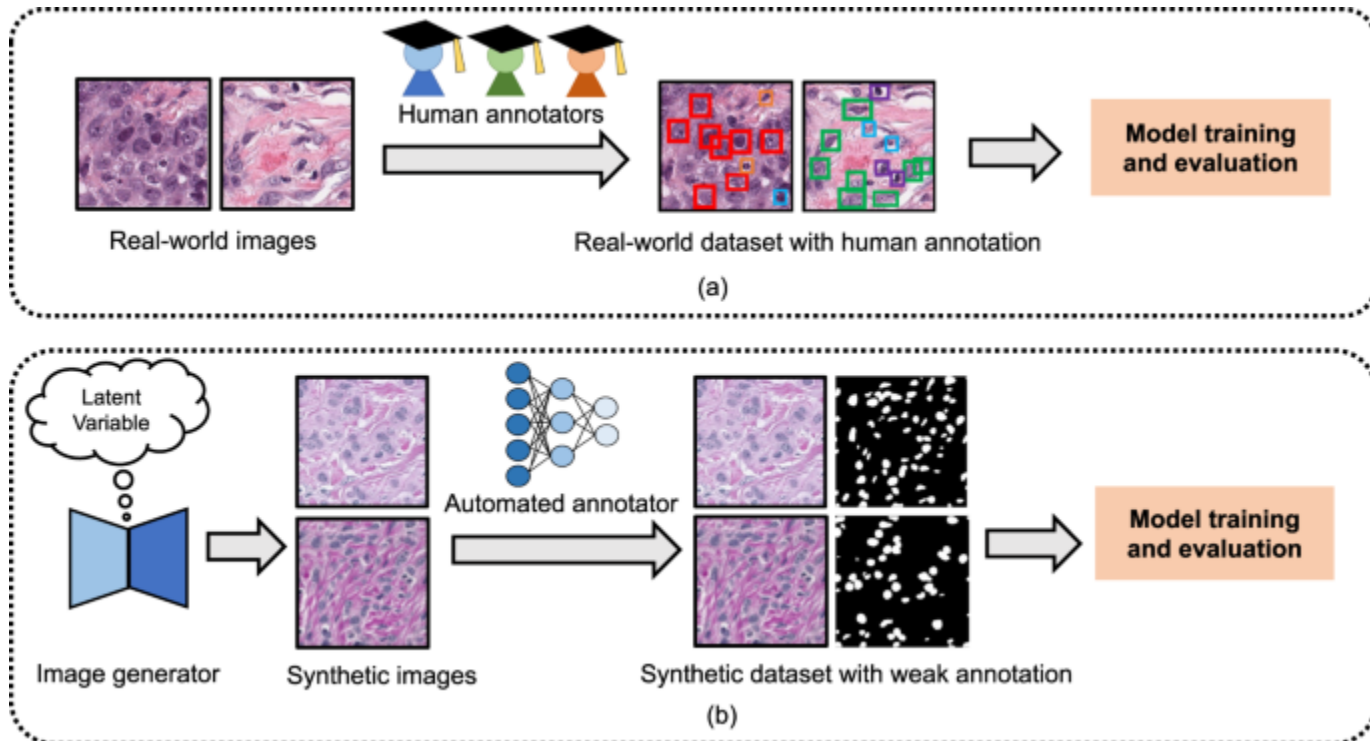


Source: Chigot et al. Style Transfer with Diffusion Models for Synthetic-to-Real Domain Adaptation. Arxiv 2025

Synth-to-Real. Bridging the Gap

- **Hybrid Training:**

- Pre-train on synthetic, fine-tune with a smaller real dataset.

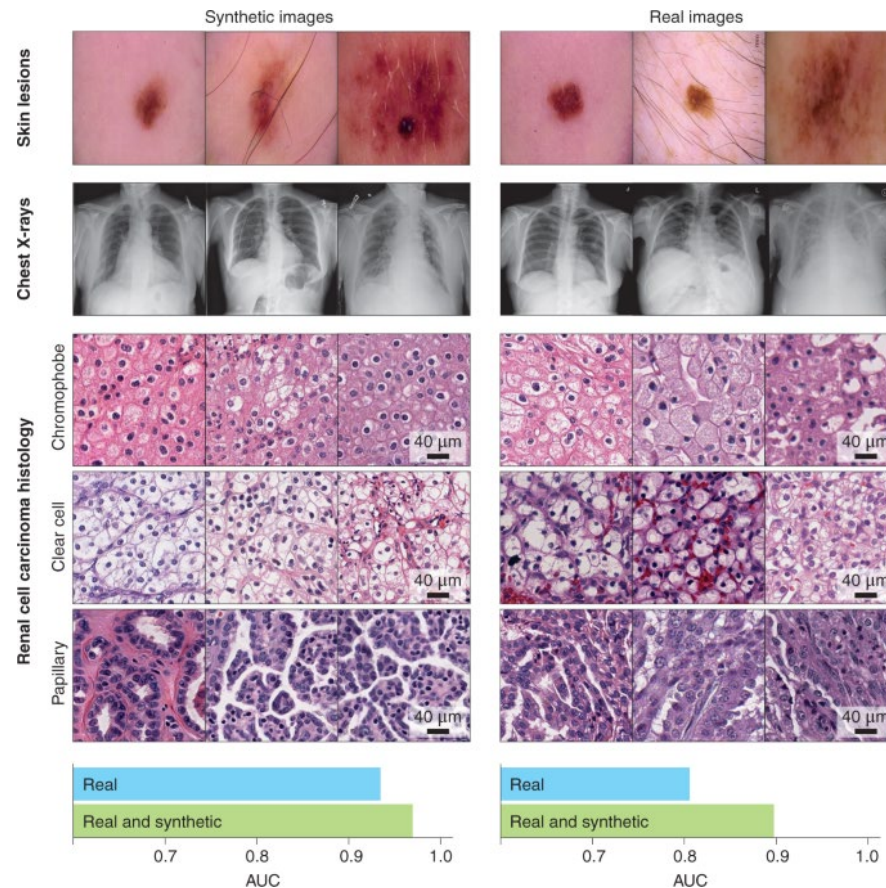


Source: © helm.ai Generative AI with Video Editing Capabilities for Autonomous Driving Development

Synth-to-Real. Bridging the Gap

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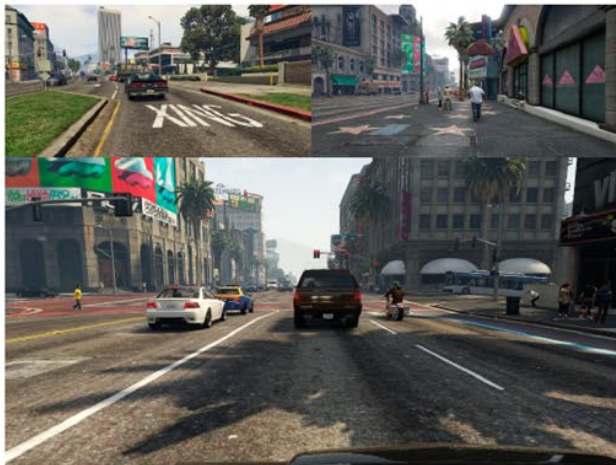
- Pre-train on synthetic, fine-tune with a smaller real dataset.



Source: "Chen et. Atl Synthetic data in machine learning for medicine and healthcare" [Nature Biomedical Engineering](#) volume 5, pages493–497 (2021)

Synth-to-Real. Bridging the Gap

- **Physics Based Rendering:**
 - Push synthetic realism closer to reality



GTAV



Synscapes



UrbanSyn

Source: urbansyn.org. Open synthetic dataset featuring photorealistic driving scenes.

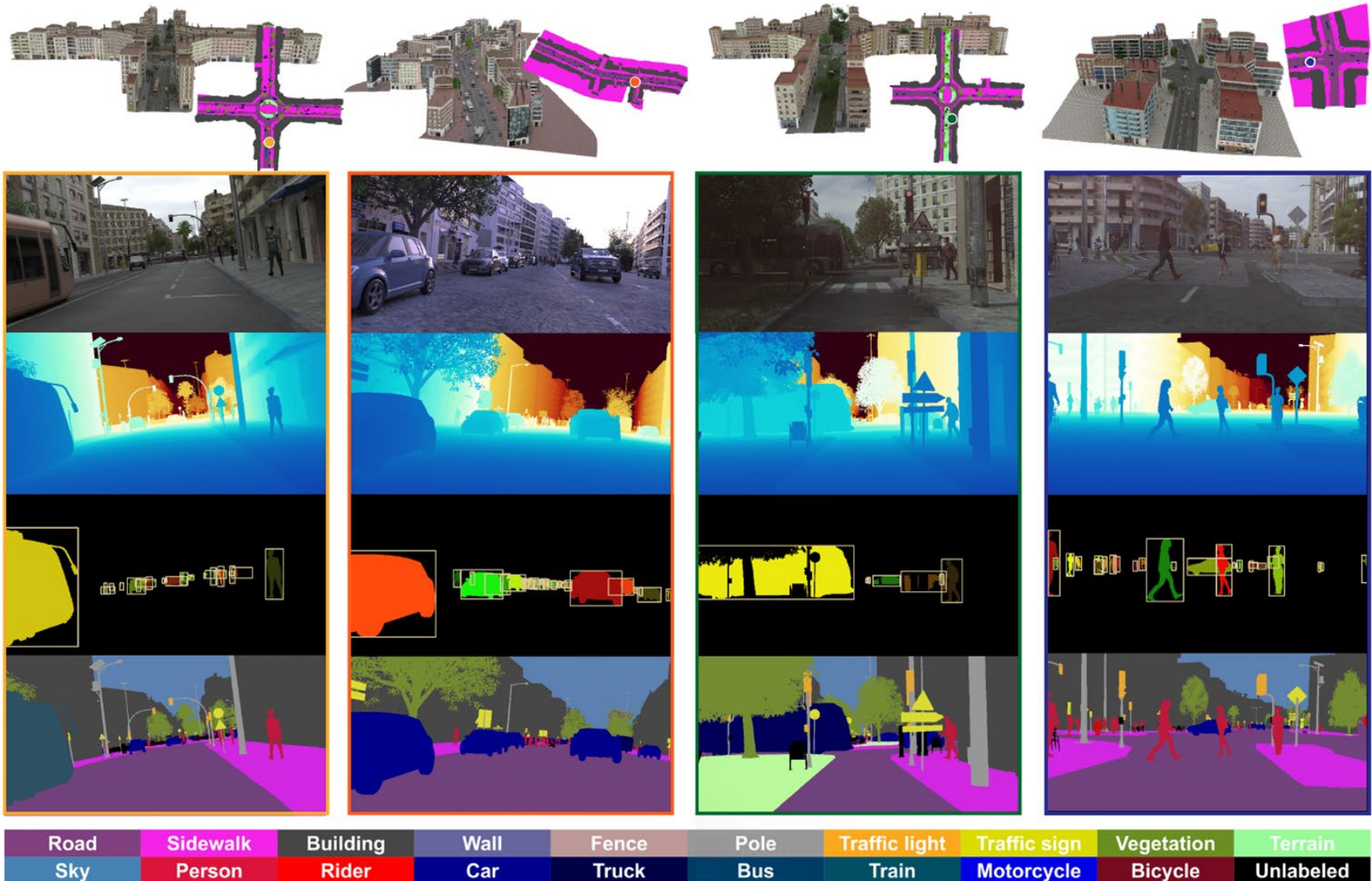
Synth-to-Real. Bridging the Gap

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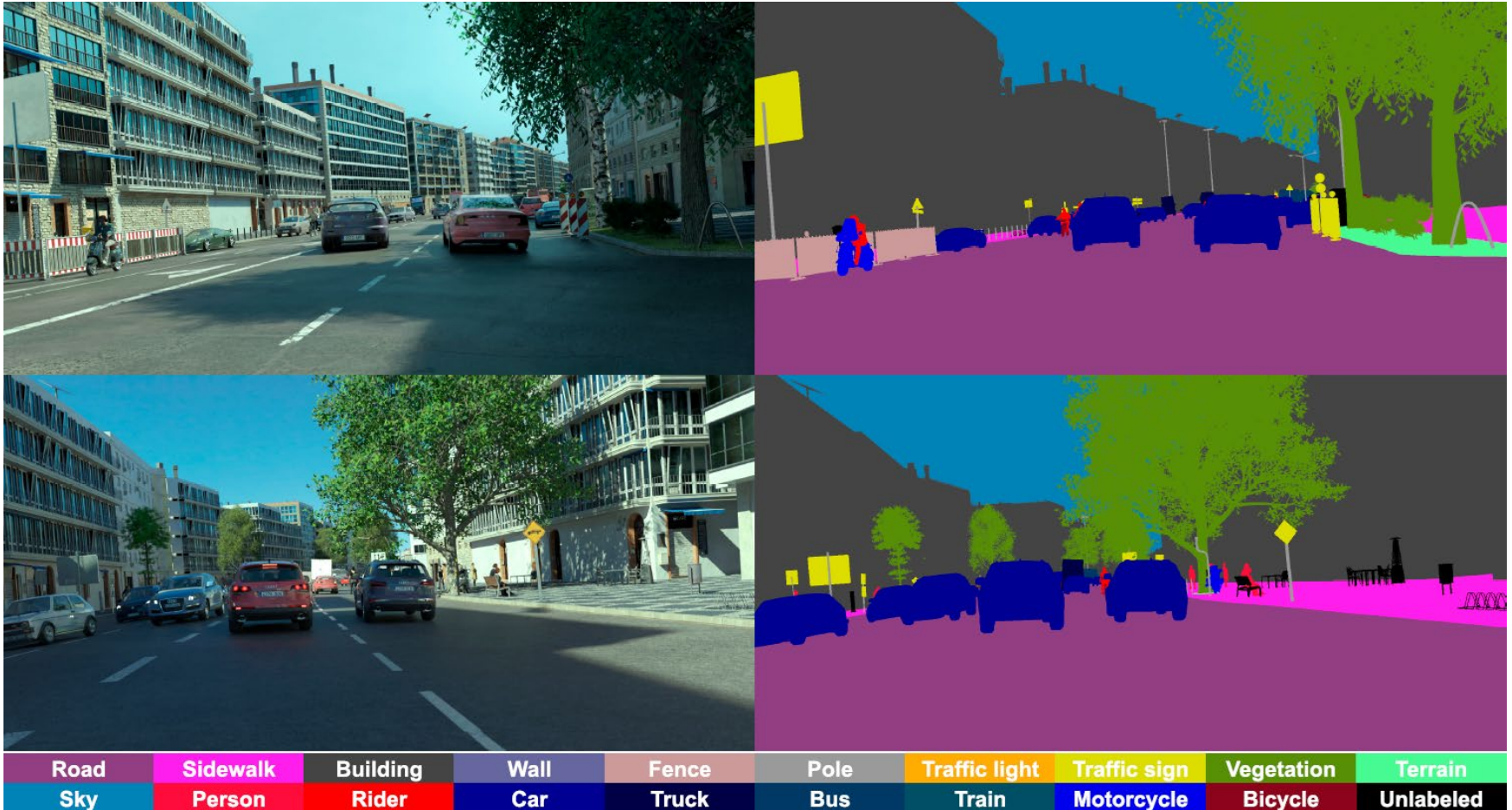
Source: urbansyn.org. Open synthetic dataset featuring photorealistic driving scenes.

UrbanSyn Dataset: Synthetic Photorealistic Driving Scenes



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UrbanSyn Dataset: Synthetic Photorealistic Driving Scenes



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UrbanSyn Dataset $\Rightarrow$ mIoU Results on Semantic Segmentation

- Synthetic Pretraining + Dataset Augmentation

averages of mIoU over three train-test runs; LAB

Target domain: Cityscapes			Target domain: BDD100K			Target domain: Mapillary Vistas		
Source domain	SegFormer	DeepLabV3+	Source domain	SegFormer	DeepLabV3+	Source domain	SegFormer	DeepLabV3+
GTAV (G)	47.8	38.0	GTAV (G)	41.2	34.0	GTAV (G)	53.3	43.0
Synscapes (S)	54.3	47.1	Synscapes (S)	37.2	27.6	Synscapes (S)	51.0	39.9
Urbansyn (U)	61.9	54.1	Urbansyn (U)	40.9	37.1	Urbansyn (U)	57.1	49.4
G + S	64.2	59.6	G + S	47.8	42.5	G + S	61.2	53.8
G + S + U	<u>70.6</u>	<u>68.2</u>	G + S + U	<u>51.7</u>	<u>45.3</u>	G + S + U	<u>67.6</u>	<u>62.4</u>
Cityscapes	81.5	79.8	BDD100K	61.5	61.2	Mapillary Vistas	79.5	77.2

G + S + U $\rightarrow$ Target						
Target domain	Baseline	HRDA ^{1,*}	Reference	Baseline	Co-Training	Reference
Cityscapes	70.6	76.2	81.5	68.2	75.3	79.8
BDD100K	51.7	59.0	61.5	45.3	53.8	61.2
Mapillary Vistas	67.6	68.1	79.5	62.4	67.9	77.2

[1] L. Hoyer, D. Dai, L. v. Gool, *HRDA: Context-aware high-resolution domain-adaptive semantic segmentation*, ECCV'2022.

(*) ImageNet feature distance disabled

Take-home messages:

- Working with strong baselines is important for UDA methods.
- Urbansyn alone provides the strongest baselines compared to GTAV and Synscapes (and SYNTHIA).
- But, even better, Urbansyn complements GTAV and Synscapes so providing the strongest baselines.
- These *Three Musketeers* allow to significantly close the synth-to-real domain gap using UDA methods.

Source: urbansyn.org. Open synthetic dataset featuring photorealistic driving scenes.

Methods for Generating Synthetic Images

- Generative AI Models (**GANs**, diffusion models, control-net)
→ good for augmentation and realism.



Source: Mures et al. Should I Render or Should AI Generate? Crafting Synthetic Semantic Segmentation Datasets With Controlled Generation, IEEE-CG&A, 2025

Real or Fake People?

- In some cases, the use of synthetic data may exploit human biases for malicious purposes, thus it also may come with some safety concerns ...



Source: thispersondoesnotexist.com (1024×1024)

Real or Fake People?

[Nightingalea and Farid, 2022]

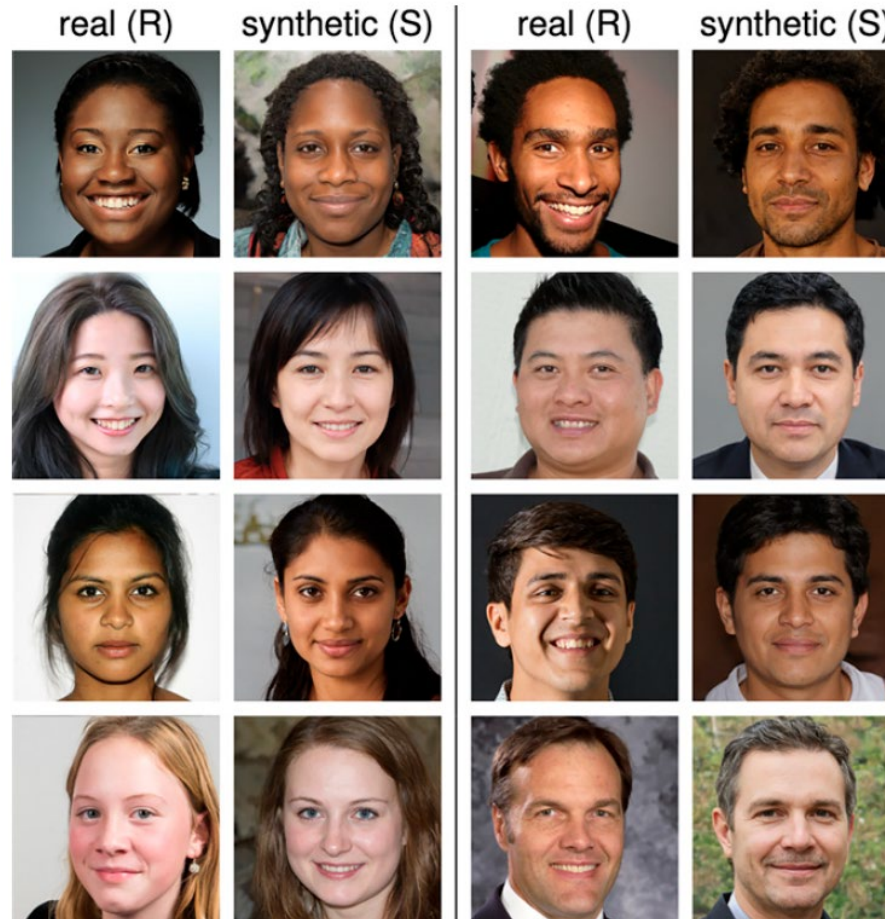


Fig. 4. A representative set of matched real and synthetic faces.

Source: Nightingalea and Farid. AI-synthesized faces are indistinguishable from real faces and more trustworthy, PNAS 2022

Which faces are more trustworthy?

- AI-synthesized faces are indistinguishable from real faces and more trustworthy.
 - 1 (less trustworthy)
 - ...
 - 7 (maximum trustworthy)

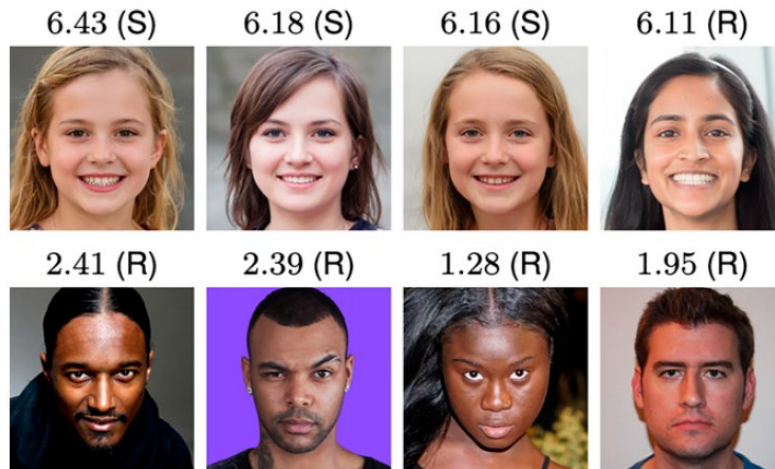


Fig. 3. The four most (*Top*) and four least (*Bottom*) trustworthy faces and their trustworthy rating on a scale of 1 (very untrustworthy) to 7 (very trustworthy). Synthetic faces (S) are, on average, more trustworthy than real faces (R).

[Nightingalea and Farid, 2022]

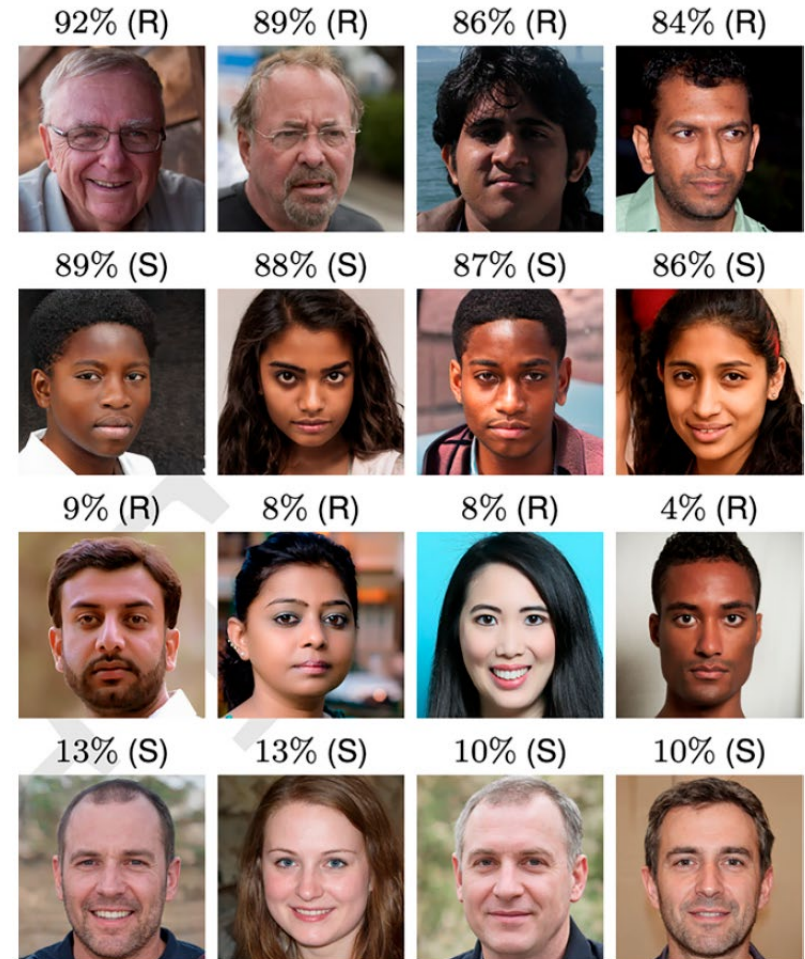
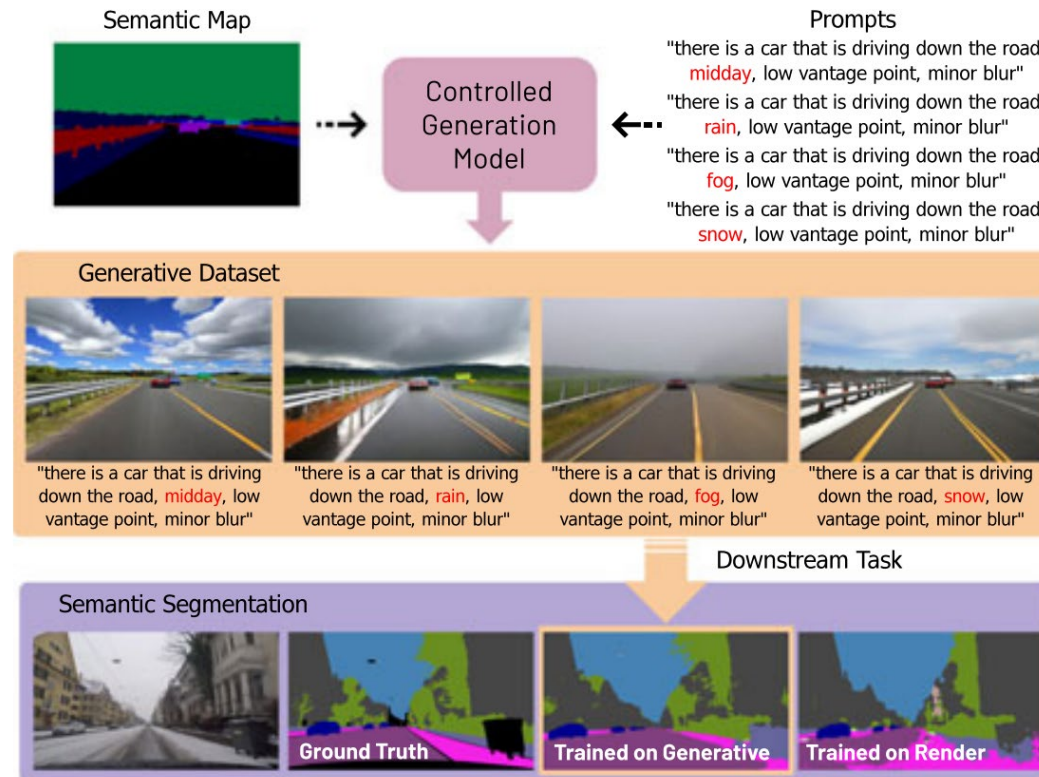


Fig. 1. The most (*Top and Upper Middle*) and least (*Bottom and Lower Middle*) accurately classified real (R) and synthetic (S) faces.

Source: Nightingalea and Farid. AI-synthesized faces are indistinguishable from real faces and more trustworthy, PNAS 2022

Methods for Generating Synthetic Images

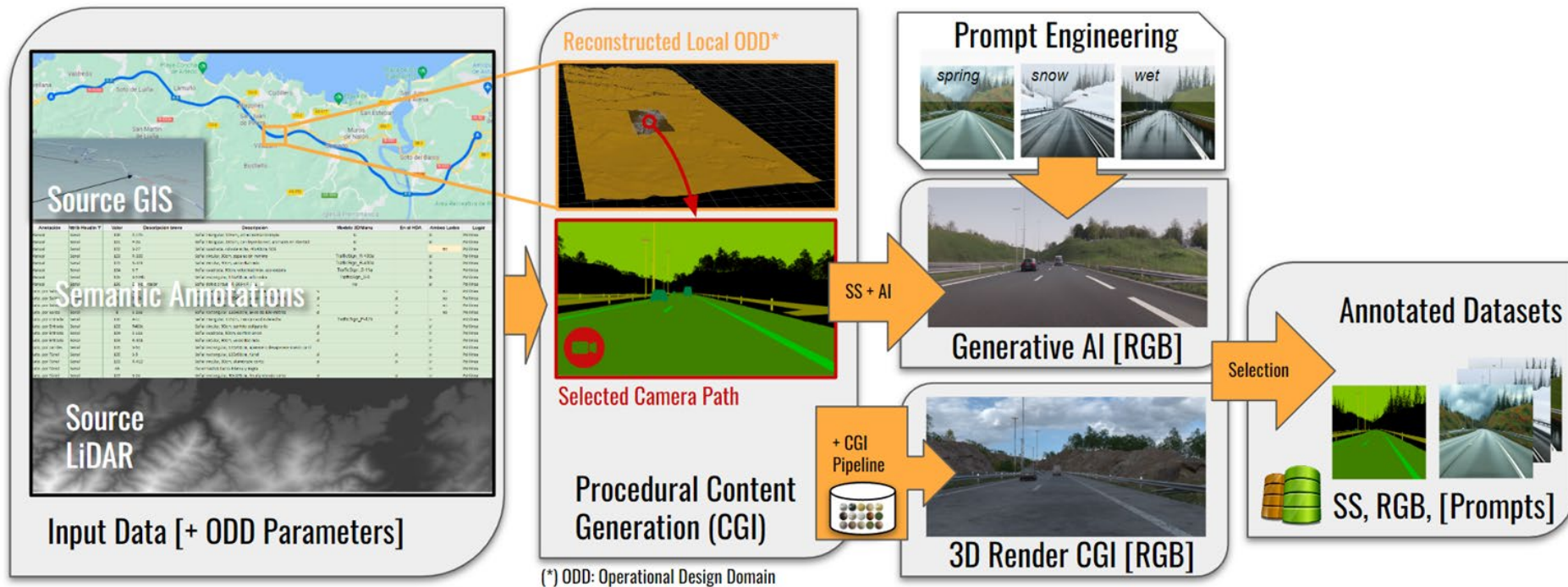
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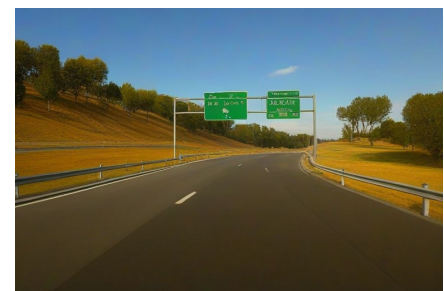
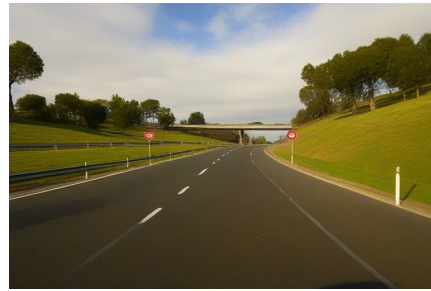
- Hybrid Methods:
 - CGI Procedural Content Generation + Generative AI synthesis ...



Source: PGAS-ADA, Googlemaps, gpsvisualizer.com, PGAS-ADA

Methods for Generating Synthetic Images

- Hybrid Methods:
 - CGI Procedural Content Generation + Generative AI synthesis ...



Source: HighwayScapes

CGI Videogame GTA-V + Generative AI (“germany-style”)



Source: GTAV reimagined by Generative AI

Synth-to-Real. The Validation Challenge

- Validation is challenging: real-world annotated datasets may still have errors.
- Examples of inconsistent annotation errors in real-world datasets
 - e.g. Cityscapes (see Figs. 8 & 9)



Fig. 8: Left to right: RGB image from Cityscapes' validation set, its semantic ground truth, and its co-training inference. Human labeling misses Traffic Light and Traffic Sign instances that co-training labels (boxed in red). In addition, a rider in the bottom right of the image is incorrectly labeled as a Pedestrian by humans, while co-training identifies it as Rider + Bicycle.



Fig. 9: Cityscapes images from the validation set containing inconsistencies in human labeling for the Fence class. Note how different criteria have been used to label the Fences in the left and right images.

Source: Gómez et al. All for one, and one for all: Urbansyn dataset, the third musketeer of synthetic driving scenes, Neurocomputing 2025.

Synthetic & Simulated Datasets. Benefits and Limitations

- Benefits:
 - They help to address limitations of real-world datasets previously mentioned
- Limitations:
 - Synthetic–real **domain gap**
 - Risk of **over-fitting** to **simulation artifacts**
 - Still need some (good) real data for **validation**
- The use of synthetic data to (re)train new models is known as ...
 - **Model collapse** for generative models
 - Model collapse is a **degenerative process** affecting generations of learned generative models, in which the generated data end up polluting the training set of next generations.
 - Being trained on polluted data, they then misperceive reality [Shumailov, 2024].
 - **Unfairness feedback** loops for supervised models [Wyllie, 2024].
 - If training data is **biased**, new data collected from the new system reflects this skew, making the bias even worse.

Practical Hints

- Practical Hints
 - Strategies for combining synthetic and real data:
 - Pre-training on synthetic → fine-tune on real
 - Use domain adaptation (e.g. CycleGAN, style transfer)
 - Best practices:
 - Match target domain conditions (resolution, noise, camera characteristics)
 - Validate with (diverse & good) real-world test sets
 - Focus on distributions/diversity, not just on image style

Summary. Q&A

- Key takeaways:
 - Lower labeling costs → automatic annotations reduce manual effort.
 - Expanded coverage → can generate rare events, balanced classes, and diverse conditions.
 - Bias mitigation → controlled data generation may address certain data imbalance.
 - Scalability & safety → simulate dangerous, costly, or impossible real-world scenarios at scale.
 - Not a silver bullet → synthetic data can fail, especially without careful validation.
- Remaining open questions:
 - Domain gap
 - How to systematically reduce the performance drop between synthetic and real data?
 - The scaling of synthetic data
 - How much synthetic data is needed, and when does “more” stop helping?
 - Licensing & copyright: licensing, copyright of assets, and responsible use.

Course Session. Summary

1. Photogrammetry and Computer Vision Fundamentals
2. A gentle introduction to Novel View Synthesis (NVS)
3. **Synthetic image data generation for AI consumption**

INDITEXUDC

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<https://catedrainditexalgoritmosverdes.com/>

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